N M EMRAN HUSSAIN

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WORK STATUS: Work permit (EAD) Issued | Available Immediately | Relocation available.

SUMMARY

Business Data Scientist and MSBA graduate specializing in Generative AI systems, customer lifetime value (CLV) modeling, and A/B testing. Proven ability to translate complex NLP pipelines and market sizing algorithms into data-backed go-to-market strategies and scalable revenue opportunities.

TECHNICAL SKILLS

- **Languages & Core Frameworks:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow, Keras, PyTorch), R (tidyverse, ggplot2), SQL
- **AI & Generative Systems:** Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformers, LangChain, Vector Databases (ChromaDB), NLP (Word2Vec, FastText, Sentiment Analysis)
- **Data Engineering & Cloud:** PySpark, Spark, PostgreSQL, ETL Pipelines, Google Cloud Platform (GCP), Git/GitHub
- **Business Strategy & Advanced Analytics:** A/B Testing, Customer Lifetime Value (CLV), Churn Optimization, Explainable AI (XAI, SHAP), ROI Analysis, ROMI Analysis, Stakeholder Management

PROFESSIONAL EXPERIENCE

Community Dreams Foundation (Skills-Based Volunteer) | Business Data Scientist | July 2026 – Present

Project: Elpsan Bio (Healthcare Analytics & Predictive Modeling)

- Modeled revenue potential for 15+ FDA drug candidates, programmatically filtering hundreds of clinical matches into 5–10 high-viability, high-ROI therapeutic targets.
- Translated NLP models into automated stakeholder dashboards, accelerating go-to-market data retrieval by 15–20% for the research team.

AMPERE LTD. | Data Analytics Lead | 2019

- Engineered Python and R regression models to forecast commercial demand, generating a 9% revenue lift and reducing overstock by 5%.
- Automated cost-tracking data pipelines to monitor pricing variances, successfully protecting 10% operating margins.

ANRHUS & SHUZOS CO. | Assistant Manager | 2009 – 2018

- Spearheaded localized data tracking workflows to optimize infrastructure logistics and accelerate asset time-to-revenue.
- Streamlined inventory analytics and audit reporting, cutting overhead expenses by 3% and boosting operational efficiency by 4%.

TECHNICAL & INDEPENDENT PROJECTS

Cardiovascular Risk Assessment & Triage Assistant | August 2026 – Present | github.com/nmemranhussain/cardio-triage-4-agents

Vertex AI, BigQuery ML, Vector Search (RAG), SHAP, Streamlit, Docker, Cloud Run

- Architected a 4-agent clinical pipeline utilizing Vertex AI Gemini to parse unstructured vitals and a BigQuery ML boosted tree to predict heart failure mortality with SHAP explainability.
- Engineered a RAG system via Vertex AI Vector Search for medical guideline retrieval, deploying the solution to Cloud Run as a serverless Streamlit dashboard with active MLOps drift monitoring.

Diabetic Readmission Risk & Clinical BI Copilot | January 2026 – August 2026 | github.com/nmemranhussain/DiaVigil

Groq API, DuckDB, XGBoost, SHAP, Plotly, Google Colab, Python

- Architected a 3-agent clinical triage application leveraging the Groq API for rapid LLM orchestration to evaluate 30-day diabetic readmission risk across 101,700+ patient records.
- Automated DuckDB SQL extraction and XGBoost feature predictions, pipelining executive BI summaries and SHAP risk drivers into interactive Plotly cohort visualizations directly within Google Colab.

Predictive-RAG Retention Engine | August 2025 – December 2025 | github.com/nmemranhussain/RAG-Retention

Gemini 2.5 Flash, Reinforcement Learning, PySpark, ChromaDB, LangChain, GCP

- Engineered an AI retention architecture utilizing LinUCB Bandits and Gemini 2.5 Flash within Google Cloud Platform.
- Deployed a Reinforcement Learning and RAG-based pipeline using PySpark, LangChain, and ChromaDB to optimize enterprise decision support and mitigate customer churn.

EDUCATION & TECHNICAL UPSKILLING

Google Cloud Professional ML Engineer (Candidate) | 2026

- Applied GCP architectural heuristics to design ML workflows utilizing BigQuery, Docker, and UV.

The George Washington University, School of Business | Washington, DC

MS in Business Analytics (MSBA) | December 2025

- *Relevant Courses:* Responsible Machine Learning, Practical Artificial Intelligence, Working with Large Dataset (Big Data), AI in Marketing, Customer Analytics, Text Analytics (NLP).

North South University, School of Business | Dhaka, Bangladesh

Master of Business Administration (MBA) | May 2023

- *Relevant Courses:* Marketing Research, Brand Management, Global Marketing, Marketing Strategy, Marketing Analytics.